

# NLP

## 2) Context Dependent

Paul is a good boy having good morals.

Paul is good in playing Cricket.

Paul is good in ?      Morality  
                                         Playing cricket

I want to meet John but he was sleeping.

context dependency

## 3) Complex Grammar

Grammar rules, Sarcasm

POS :- English Language

8 POS :-      Noun  
                 Pronoun      Interjection  
                 Verb  
                 Adverb  
                 Preposition  
                 Conjunction  
                 Adjective

## 4) Ever growing Vocabulary

NLP :- Textual Data

### Classical NLP Pipeline

Text Preprocessing  $\longrightarrow$  Text Representation  $\longrightarrow$  NLP Task

\* Numbers = 1, 2, 3

\* Vectors = [1.01, 1.02]

\* Tensors = Vectors on GPU

### Text Processing

1) Tokenization

2) Stemming & Lemmatization

Loss of Information

Stem \* playing :- play \* studies

played :- play study

lemma better :- good  
best :- good

3) Stopwords Removal  
in, at, the

4) Normalization

Mix:- Upper Cases & Lower Cases

\* ~~✓~~ Lower Cases \*

Text Representation

1) Embedding

(i) OHE

